

Predictive Modeling and Risk Scoring for Bank Customer Churn

Abstract

Customer churn is an important business problem for retail banks because losing customers can reduce customer lifetime value, revenue stability, cross-selling opportunities and long-term competitiveness. This study develops a machine-learning based churn prediction and risk-scoring framework using a 10,000-record European bank customer dataset. The workflow includes data preparation, categorical encoding, numerical scaling, feature engineering, stratified train-test splitting and comparison of Logistic Regression, Decision Tree, Random Forest and Gradient Boosting models.

1. Introduction

Bank customer retention requires early identification of customers who may leave the institution. Predictive modeling can estimate churn probability for an individual customer and support targeted retention decisions.

2. Problem Statement

The objective is to build a predictive system that estimates whether a bank customer is likely to churn and produces a quantitative churn probability for retention decision support.

3. Objectives

Predict churn, generate probability scores, identify churn drivers, improve interpretability, enable scenario-based risk analysis and provide a live dashboard.

4. Dataset Description

The supplied dataset contains 10,000 customer records and 14 columns. The target is Exited. CustomerId and Surname are treated as identifiers and excluded from model training.

5. Methodology

The workflow includes missing-value handling, one-hot encoding for Geography and Gender, numerical scaling, feature engineering, and a stratified 80/20 train-test split. Engineered features are Balance-to-Salary ratio, Product Density, Product Engagement interaction and Age-Tenure interaction.

6. Models

Logistic Regression is used as an interpretable baseline. Decision Tree, Random Forest and Gradient Boosting provide tree-based and ensemble alternatives.

7. Model Evaluation

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Gradient Boosting	86.75%	78.86%	47.67%	59.42%	86.22%
Decision Tree	76.35%	45.26%	77.40%	57.12%	83.81%
Random Forest	77.10%	45.91%	70.27%	55.53%	82.25%
Logistic Regression	72.60%	39.49%	65.11%	49.17%	75.40%

Gradient Boosting achieved the highest ROC-AUC and accuracy in the included experiment. Decision Tree achieved the highest recall, which can be useful when missing a potential churner is more costly.

8. Risk Scoring Application

The Streamlit application accepts customer characteristics and returns churn probability, a Low/Medium/High risk category and a churn flag using a 0.50 probability threshold.

9. Business Use

Risk scores can prioritize retention campaigns, service reviews, customer engagement and relationship-manager effort.

10. Limitations

The results depend on the supplied dataset and selected model configuration. The data does not contain every behavioral, transaction-level or service-quality variable, so scores should be treated as decision-support outputs.

11. Future Scope

Future work can include hyperparameter tuning, cross-validation, probability calibration, SHAP explanations, business-cost threshold optimization, time-based validation, CRM integration and model-drift monitoring.

12. Conclusion

The project demonstrates an end-to-end machine-learning workflow for bank customer churn prediction and risk scoring. The model and dashboard provide a practical foundation for proactive retention and data-driven customer strategy.